```
# normal distribution==>data mean=0,standard deviation=1
# central tendency(mean,median,mode)=>left side and right side equal data point
```

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('/content/insurance.csv')
```

```
df.head(2)
```

| | age | sex | bmi | children | smoker | region | charges |
|---|---|---|---|---|---|---|---|
| **0** | 19 | female | 27.90 | 0 | yes | southwest | 16884.9240 |
| **1** | 18 | male | 33.77 | 1 | no | southeast | 1725.5523 |

Next steps: Generate code with df | New interactive sheet

```
df['sex']=df['sex'].map({'female':0,'male':1})
```

```
df.isnull().sum()
```

| | 0 |
|---|---|
| **age** | 0 |
| **sex** | 0 |
| **bmi** | 0 |
| **children** | 0 |
| **smoker** | 0 |
| **region** | 0 |
| **charges** | 0 |

**dtype:** int64

```
df['smoker']=df['smoker'].map({'yes':1,'no':0})
```

```
df['region']=df['region'].map({'southwest':1,'southeast':2,'northwest':3,'northeast':4})
```

```
df.sample(4)
```

| | age | sex | bmi | children | smoker | region | charges |
|---|---|---|---|---|---|---|---|
| **226** | 28 | 1 | 38.06 | 0 | 0 | 2 | 2689.4954 |
| **279** | 51 | 0 | 21.56 | 1 | 0 | 2 | 9855.1314 |
| **864** | 51 | 1 | 25.40 | 0 | 0 | 1 | 8782.4690 |
| **246** | 60 | 0 | 38.06 | 0 | 0 | 2 | 12648.7034 |

```
x=df.drop(columns=['charges'])
y=df['charges']
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=1)
```

```
np.round(x_train.describe(),2)
```

| | age | sex | bmi | children | smoker | region |
|---|---|---|---|---|---|---|
| **count** | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 |
| **mean** | 39.00 | 0.51 | 30.67 | 1.10 | 0.21 | 2.49 |
| **std** | 13.97 | 0.50 | 6.13 | 1.21 | 0.40 | 1.11 |
| **min** | 18.00 | 0.00 | 15.96 | 0.00 | 0.00 | 1.00 |
| **25%** | 27.00 | 0.00 | 26.30 | 0.00 | 0.00 | 2.00 |
| **50%** | 39.00 | 1.00 | 30.50 | 1.00 | 0.00 | 2.00 |
| **75%** | 51.00 | 1.00 | 34.66 | 2.00 | 0.00 | 3.00 |
| **max** | 64.00 | 1.00 | 53.13 | 5.00 | 1.00 | 4.00 |

```
from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()
```

```
x_train_sc = sc.fit_transform(x_train) #fit means learn the parameter and transform mean apply on the data
```

```
x_train_new = pd.DataFrame(x_train_sc,columns=x_train.columns)
```

```
np.round(x_train_new.describe(),2)
```

| | age | sex | bmi | children | smoker | region |
|---|---|---|---|---|---|---|
| **count** | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 |
| **mean** | -0.00 | -0.00 | -0.00 | -0.00 | 0.00 | 0.00 |
| **std** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| **min** | -1.50 | -1.02 | -2.40 | -0.91 | -0.51 | -1.33 |
| **25%** | -0.86 | -1.02 | -0.71 | -0.91 | -0.51 | -0.44 |
| **50%** | -0.00 | 0.98 | -0.03 | -0.08 | -0.51 | -0.44 |
| **75%** | 0.86 | 0.98 | 0.65 | 0.75 | -0.51 | 0.46 |
| **max** | 1.79 | 0.98 | 3.67 | 3.23 | 1.97 | 1.36 |

```
# task
# covid data set ==>handle missing data,convert in num,data dividation,normal distribution(mean=0,std=1)
```

```
# normlization(min_value=0,max_value=1)
```

```
from sklearn.preprocessing import MinMaxScaler
```

```
mn=MinMaxScaler()
```

```
x_train_mn=mn.fit_transform(x_train)
```

```
x_train_new=pd.DataFrame(x_train_mn,columns=x_train.columns)
```

```
np.round(x_train_new.describe(),2)
```

| | age | sex | bmi | children | smoker | region |
|---|---|---|---|---|---|---|
| **count** | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 | 1070.00 |
| **mean** | 0.46 | 0.51 | 0.40 | 0.22 | 0.21 | 0.50 |
| **std** | 0.30 | 0.50 | 0.16 | 0.24 | 0.40 | 0.37 |
| **min** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **25%** | 0.20 | 0.00 | 0.28 | 0.00 | 0.00 | 0.33 |
| **50%** | 0.46 | 1.00 | 0.39 | 0.20 | 0.00 | 0.33 |
| **75%** | 0.72 | 1.00 | 0.50 | 0.40 | 0.00 | 0.67 |
| **max** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

Start coding or generate with AI.

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